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D. Richard Kuhn

Explainability, Verification, Validation, and Assurance of Security in AI-enabled Systems - D. Richard Kuhn, M S Raunak, Sanjay Rekhi, NIST Computer Security Division 773.02 Autonomous systems are increasingly seen in security-critical domains, such as industrial control systems and autonomous aircraft. Unfortunately, methods developed for ultra-reliable software, such as avionics, depend on measures of structural coverage that do not apply to neural networks or other black-box functions often used in machine learning. One approach that can be used is to ensure that all relevant combinations of input values have been tested and verified for correct operation. Combinatorial coverage measures provide an efficient means of achieving this type of verification, and validating it in real-world use. Artificial intelligence and machine learning (AI/ML) systems often equal or surpass human performance in applications ranging from medical systems to self-driving cars, and defense. But ultimately a human must take responsibility, so it is essential to be able to justify the system's action or decision. What combinations of factors support the decision? Why was another action not taken? How do we know the system is working correctly and securely? We consider explainability to be part of the larger problem of verification and validation for autonomous systems and artificial intelligence. Measurement-based test methods and tools for ensuring security and reliability of autonomous systems must address both verification and validation. Verification Input space model measurement – Verification means ensuring that the system behaves as specified for all inputs. This requires ensuring that training and test data closely match the environment for use, and rare combinations are included in autonomous systems training and testing. We can apply covering arrays for all t-way (e.g., all triples of values) combinations of parameter values, or measure the coverage of tests that are applied. See link [1] below for background and [2] for case studies where covering arrays have been applied to autonomous vehicle testing. 1. https://csrc.nist.gov/Projects/automated-combinatorial-testing-for-software/coverage-measurement 2. https://csrc.nist.gov/Projects/automated-combinatorial-testing-for-software/case-studies-and-examples/autonomous-vehicles Validation Explainability – Validation means ensuring that the system meets the needs of the user including its security and trustworthiness. If we cannot explain or justify decisions of an AI application, then it is difficult to trust the system. Even black-box components such as neural nets can be hard to trust based only on a track record of use, as these systems are "brittle", in the sense that small changes can result in enormous errors, such as adversarial imaging where a stop sign is interpreted as a speed limit sign. Combinatorial methods allow us to produce explanations or justifications of decisions in AI/ML systems. Explainability is a necessary but not sufficient condition for assurance in these systems. Explainability is key in both using and assuring security and reliability for autonomous systems and other applications of AI and machine learning. Secure AI-enabled systems will require measurement methods to ensure that the training and testing data for AI adequately reflect the environment in which the system will be used.

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One approach that can be used is to ensure that all relevant *combinations of input values* have been tested and verified for correct operation. Combinatorial coverage measures provide an efficient means of achieving this type of verification, and validating it in real-world use.

Artificial intelligence and machine learning (AI/ML) systems often equal or surpass human performance in applications ranging from medical systems to self-driving cars, and defense. But ultimately a human must take responsibility, so it is essential to be able to justify the system's action or decision. What combinations of factors support the decision? Why was another action not taken? How do we know the system is working correctly and securely? We consider explainability to be part of the larger problem of verification and validation for autonomous systems and artificial intelligence.

Measurement-based test methods and tools for ensuring security and reliability of autonomous systems must address both verification and validation.

Verification

Input space model measurement – Verification means ensuring that the system behaves as specified for all inputs. This requires ensuring that training and test data closely match the environment for use, and rare combinations are included in autonomous systems training and testing. We can apply covering arrays for all *t*-way (e.g., all triples of values) combinations of parameter values, or measure the coverage of tests that are applied. See link [1] below for background and [2] for case studies where covering arrays have been applied to autonomous vehicle testing.

1. https://csrc.nist.gov/Projects/automated-combinatorial-testing-for-software/coverage-measurement
2. https://csrc.nist.gov/Projects/automated-combinatorial-testing-for-software/case-studies-and-examples/autonomous-vehicles

Validation

Explainability – Validation means ensuring that the system meets the needs of the user including its security and trustworthiness. If we cannot explain or justify decisions of an AI application, then it is difficult to trust the system. Even black-box components such as neural nets can be hard to trust based only on a track record of use, as these systems are "brittle", in the sense that small changes can result in enormous errors, such as adversarial imaging where a stop sign is interpreted as a speed limit sign.

Combinatorial methods allow us to produce explanations or justifications of decisions in AI/ML systems. Explainability is a necessary but not sufficient condition for assurance in these systems. Explainability is key in both using and assuring security and reliability for autonomous systems and other applications of AI and machine learning.

Secure AI-enabled systems will require measurement methods to ensure that the training and testing data for AI adequately reflect the environment in which the system will be used.